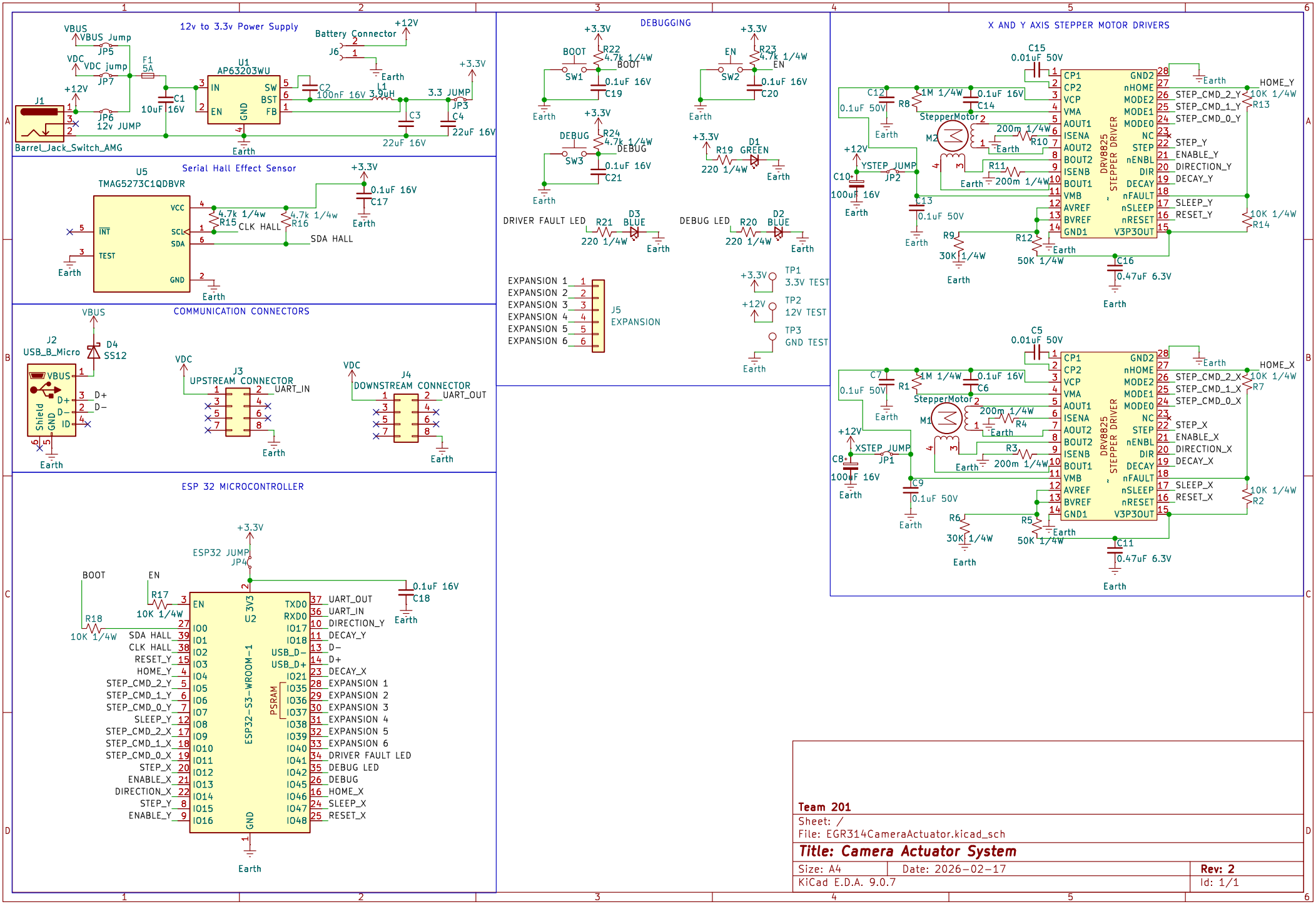

12v to 3.3v Power Supply
VBUS
VBUS_Jump
JP5
VDC
VDC_jump
JP7
+12V
JP6
12v JUMP
J1
Barrel_Jack_Switch_AMG
F1
5A
C1
10uF 16V
U1
AP63203WU
IN
EN
SW
BST
FB
GND
C2
100nF 16V
L1
3.9uH
C3
22uF 16V
C4
22uF 16V
3.3 JUMP
JP3
+3.3V
Battery Connector
J6
Earth
Serial Hall Effect Sensor
U5
TMAG5273C1QDBVR
VCC
INT
SCL
SDA
TEST
GND
4.7k 1/4w
R15
CLK HALL
4.7k 1/4w
R16
SDA HALL
C17
0.1uF 16V
COMMUNICATION CONNECTORS
J2
USB_B_Micro
D4
SS12
VBUS
D+
D−
ID
GND
Shield
J3
UPSTREAM CONNECTOR
UART_IN
J4
DOWNSTREAM CONNECTOR
UART_OUT
ESP 32 MICROCONTROLLER
ESP32 JUMP
JP4
C18
0.1uF 16V
R17
10K 1/4w
R18
10K 1/4w
U2
ESP32-S3-WROOM-1
3V3
GND
PSRAM
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
USB_D−
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
TXD0
RXD0
SDA HALL
CLK HALL
RESET_Y
HOME_Y
STEP_CMD_2_Y
STEP_CMD_1_Y
STEP_CMD_0_Y
SLEEP_Y
STEP_CMD_2_X
STEP_CMD_1_X
STEP_CMD_0_X
STEP_X
ENABLE_X
DIRECTION_X
STEP_Y
ENABLE_Y
UART_OUT
UART_IN
DIRECTION_Y
DECAY_Y
D−
D+
DECAY_X
EXPANSION 1
EXPANSION 2
EXPANSION 3
EXPANSION 4
EXPANSION 5
EXPANSION 6
DRIVER FAULT LED
DEBUG LED
HOME_X
SLEEP_X
RESET_X
DEBUGGING
BOOT
SW1
R22
4.7k 1/4W
0.1uF 16V
C19
EN
SW2
R23
4.7k 1/4W
C20
DEBUG
SW3
R24
4.7k 1/4W
C21
R19
220 1/4W
D1
GREEN
DRIVER FAULT LED
R21
220 1/4W
D3
BLUE
DEBUG LED
R20
220 1/4W
D2
BLUE
J5
EXPANSION
TP1
3.3V TEST
TP2
12V TEST
TP3
GND TEST
X AND Y AXIS STEPPER MOTOR DRIVERS
C15
0.01uF 50V
C12
0.1uF 50V
R8
1M 1/4W
C14
0.1uF 16V
StepperMotor
M2
200m 1/4W
R10
R11
200m 1/4W
YSTEP_JUMP
JP2
C10
100uF 16V
C13
0.1uF 50V
R9
30K 1/4W
R12
50K 1/4W
C16
0.47uF 6.3V
DRV8825
STEPPER DRIVER
CP1
CP2
VCP
VMA
AOUT1
ISENA
AOUT2
BOUT2
ISENB
BOUT1
VMB
AVREF
BVREF
GND1
GND2
nHOME
MODE2
MODE1
MODE0
NC
STEP
nENBL
DIR
DECAY
nFAULT
nSLEEP
nRESET
V3P3OUT
HOME_Y
STEP_CMD_2_Y
STEP_CMD_1_Y
STEP_CMD_0_Y
STEP_Y
ENABLE_Y
DIRECTION_Y
DECAY_Y
SLEEP_Y
RESET_Y
10K 1/4W
R13
10K 1/4W
R14
C5
0.01uF 50V
C7
0.1uF 50V
R1
1M 1/4W
C6
0.1uF 16V
StepperMotor
M1
200m 1/4W
R4
R3
200m 1/4W
XSTEP_JUMP
JP1
C8
100uF 16V
C9
0.1uF 50V
R6
30K 1/4W
R5
50K 1/4W
C11
0.47uF 6.3V
HOME_X
STEP_CMD_2_X
STEP_CMD_1_X
STEP_CMD_0_X
STEP_X
ENABLE_X
DIRECTION_X
DECAY_X
SLEEP_X
RESET_X
10K 1/4W
R7
10K 1/4W
R2
Earth
Team 201
Sheet: /
File: EGR314CameraActuator.kicad_sch
Title: Camera Actuator System
Size: A4
Date: 2026-02-17
Rev: 2
KiCad E.D.A. 9.0.7
Id: 1/1